

Data Management – AI Failures Museum

1. Introduction

This document is an demonstration of how the data is hanled in our project.

Our library uses **manually created cases**, created by the Data/ML Lead, rather than relying on external datasets or APIs. Due to this our data mangemnt aproach is more scalability and reliability centered.

2. Data Sources

The system uses a single data source:

2.1 Internal Case Data

- **Description:**
A collection of AI failure cases created for the museum.
- **Type:**
 - Text , case descriptions and links
 - Images, ‘Artifacts’ images generated by Gemini nano Banana 2
 - Quiz content ,questions and answers
- **Origin:**
 - Created manually by the Data/ML Lead
 - Some cases are fictional, others are based on real-world events with references
- **Volume:** 11 original cases in sprint 1 , 20 total 5(non-fictinol)

No external APIs or third-party datasets are used in the system.

3. Data Architecture & Flow

We have two flows; a Backend-driven and a Front-end user centric flow

Original flow(Backend);

- Case created in New_Cases.md
- Copied over into Json file
- It pulls the Json into a selection of the object
- Each object is displayed with the html template

New flow (Frontend);

- Log into website as admin
- Select create new exhibit and can create directly in the website (which automatically carries out the backend flow)

The introduction of the **Role-Based Access Control (RBAC)** system allows:

- Curators/Admins to create/edit exhibits via the frontend
- Users to view content only

4. Data Storage

- Data is stored within the web application backend
- Likely structure includes:
 - Case title
 - Description
 - Category
 - References
 - Quizes
 - Images

Storage Characteristics

- Persistent storage
- No large-scale or real-time data ingestion
- Designed for Scalability via the frontend (post-RBAC)

5. Data Processing

Minimal processing is required due to the nature of the data:

- Manual creation ensures data is pre-cleaned
- Formatting applied for consistent display
- Quiz data structured for user interaction
- No email addresses are stored or any user data outside of the sessions

Validation

- Content reviewed before publishing
- RBAC ensures only authorised users can modify data

6. Data Privacy & Security

- User names and passwords are collected along with any comments
- System operates primarily as a **read-only content platform** for non admin roles

Security Measures

- User information is salted and hashed with sha256
 - Securing the information from decryption
- Role-Based Access Control (RBAC):
 - **User:** View only
 - **Curator:** Add/edit exhibits
 - **Admin:** Full control
- Prevents:
 - Unauthorised edits
 - Accidental data loss

7. Data Lifecycle Management

Creation

- Cases created manually by Data Lead

Storage

- Stored within the system

Updates

- Enabled via frontend editing (RBAC system)

Deletion

- Managed by Admin if required

Retention

- Data is retained indefinitely unless manually removed
- Username ,password and comments are kept

8. Failure Handling

Given the simplicity of the data system:

- Missing data , an exhibit can not be created if there are any null values.
- Corrupt data, prevented through controlled editing access
- No dependency on external APIs ,so no external failure risks

9. Dependencies & Data Inventory

- No external datasets or APIs used
- All data is internally generated

This aligns with the:

- software_data_inventory_final.xlsx

- licence_decision_final.pdf

Assets

- AI-generated images (referenced appropriately)
- Text content created by team members

10. Maintenance & Future Updates

The system is designed for easy extension:

Adding New Cases

- Use curator/admin interface front end
- **Updating Existing Data**

Done directly in the frontend

Extending System

- New categories or exhibit can be added

11. Limitations

- Due to every part of exhibit needing to be filled out so if there's an section that there no info for it would have to be filled out with a place holder.
- Potential bias in case selection
- Reliance on manual content creation limits scalability
- Risk of misinformation from fictional cases – so i added an disclaimer for the fictional cases